

Analysis of existing AI medical systems

Nicolas Gabriel Fernandez Quiroz

Team TheParents

Florida State University

CEN4090L: Software Engineering Capstone Project

Professor David A. Gaitros

Friday February 27th, 2026

Abstract

Artificial intelligence (AI) has impacted the medical field with a variety of tools and services such as virtual assistants (VAs). As AMI (Artificial Medical Intelligence) is under development, there must be an analysis of the past through how AI was implemented in the medical field, how language models perform against large language models, and what are the issues that can come up with AMI as an AI. By analyzing open-source language models (SLMs), there can be further considerations that can be found before designing AMI. According to some sources, locally run SLMs run poorly against large language models (LLMs) such as Gemini because of the large gap of computing power. That is why the design and implementation of AMI needs to be carefully considered to focus on just topics of personalized medicine to avoid hallucinations. The review of SLMs show potential templates that AMI can use for the design and implementation of the architecture and framework.

Keywords: AI (Artificial Intelligence), ML (Machine Learning), AMI (Artificial Medical Intelligence), LLM (Large Language Model), LMM (Large Multimodal Model), SLM (open-Source Language Model), LM (Language Model), VA (Virtual Assistant).

Analysis of existing AI medical systems

AI has become an increasingly innovative tool in the medical sphere with AI helping with documentation, diagnostics, triage, and much more that can be automated in healthcare. With the production of the AMI (Artificial Medical Intelligence) software product underway, this paper analyses existing AI medical systems on what they do and how they function in the current medical climate. AMI is intended to be a VA (virtual assistant) for doctors, patients, and commonfolk for topics related to personalized medicine. A virtual assistant is an AI that is designed as an LLM that takes questions or requests as a prompt (or series of prompts) and produces an answer through a response. There has been concern that these VAs hallucinate and create responses that are false claims not supported by the scientific community, or that do not have any evidence to support. The objective of AMI is to proactively combat this issue by directly providing the sources/references used to generate the response. This can help the users verify that response is valid and can give them an opportunity to do further reading on the topic if they wish to do so. Reviewing the current AI medical systems will provide an insight of the potential challenges when designing, developing, and testing AMI.

Methodology

To understand how current AI medical systems work, there will be a literature review on how AI is implemented into modern medical systems, an analysis of open-source AI healthcare VAs to see how they work. The open-source AI models that will be analyzed are: MedGemma (from Google), BioMistral-7B (from Mistral AI), LLaVA-Med (from Microsoft). There was a consideration of reviewing closed-source AI healthcare VAs such as: Olivia (from Tempus), Health (from Ada), Amazon One Medical Health (from Amazon), Zoe (from Healthee). All of these AI models were not easily accessible like ChatGPT or Gemini as the majority required a request to try out a demo of these models.

Amazon One Medical Health was out of scope for the expectations of AMI as it focused on reviewing medical records, hospital appointments, and medication management. AMI is currently being designed in a conversational (Q&A style) format without needing any sensitive details such as EHRs (Electronic Health Records) from users so analyzing this AI model would not bring anything substantial.

Zoe requires that you provide company details to use the demo from Healthee which is not currently possible as AMI is being produced by FSU students. The goal for AMI is that it is available without required login nor request details. It is not considered in the analysis as the demo could not be accessed because of the request-based entry.

Similar to Amazon's AI model, Olivia also requires you to import your health data to use it which is why it is not considered in this analysis.

Health was an interesting candidate as it is focused exclusively on symptom assessment. Besides only being a mobile-based app that requires the user to sign up and provide preliminary

data (age, biological sex, etc.), Health only provides part of what AMI is intended to do.

Experience with the application showed that you can only provide a single prompt before getting a barrage of questions before getting a report. The questions after the prompt only allowed a multiple choice of three answers (yes, no, and i don't know) with an additional optional textbox at the bottom providing a 1 sentence explanation why the question was being asked. This part of Health is not very useful to the goals of AMI because it does not provide opportunity for any additional follow-up prompts to question, the textbox explanation does not provide a source on why it is asked and the 1 sentence explanation is mostly lackluster at best. The other half of Health is the assessment report at the end of all the questions. The AI provides a list of conditions that could be causing the symptom presented in the initial prompt and gives the option for further reading on each condition. The further reading option does not provide any sources on the information provided, which is what AMI strives to solve. Upon completing the symptom assessment, there was no available history in viewing previous assessments. Overall, Health lacks many of the qualities that AMI is intended to function, so an analysis on this AI model would not be useful.

Other open-source AI models that were considered were MedSigLIP (from Google) and ClinicalBERT (from Google). MedSigLIP is designed to receive images as input and not text. ClinicalBERT on the other hand is primarily designed for analyzing EHRs but can be formatted to receive text prompts. As both AI models are not made to support text prompts, they are excluded from the analysis.

Literature Review

The rise of Generative AI and LLMs during the pandemic in 2020 gave the healthcare industry expectations on how it was going to be implemented into the space. Bajwa et. al. (2021) describes how the pandemic caused “supply-and-demand challenges” that AI could be part of the solution as technological innovation allowed healthcare to becoming increasingly digitized, with AI being able to provide healthcare delivery online (“Introduction” section). AI in 2020 to 2021 was just being used to automate repetitive and/or time-consuming tasks, with expectations that it would evolve from being able to analyze unlabeled medical data to fully being able to personalize medicine for patients (Bajwa et. al., 2021, “What are the current and future use cases of AI in healthcare?” section). It was expected that AI VAs would be embedded into all electronics (phones, watches, etc.) that could track and manage medical data (Bajwa et. al., 2021, “What are the current and future use cases of AI in healthcare?” section).

As AI VAs were being developed, multiple SLMs (open-Source Language Models) were produced so people could start using or develop their own AI models. Magnini et. al. (2025) discusses how there are concerns with reliability of these AIs and privacy in terms of sensitive health data that can be misused or leaked if not managed properly (“Abstract” section). Some locally run SLMs were tested against Gemini Pro 1.5 to see how they perform in “[i]ntent recognition”, “[d]ata extraction” and “[g]eneral response generation” (Magnini et. al., 2025, “3.1. Tasks & datasets” section). The results showed that the SLMs performance poorly against Gemini Pro 1.5, as they only reached up to 77% accuracy compared to Gemini Pro’s 88%-91% in all three categories (Magnini et. al., 2025, “4. Results” section). Further analysis showed that locally run SLMs are not a viable alternative to LLMs in terms of chatbot functionality as there is not enough computing power to avoid hallucinations (Magnini et. al., 2025, “5. Discussion &

Conclusion” section). This shows a potential roadblock with AMI that must be managed by focusing on just topics on personalized medicine.

Hallucinations are not just the only roadblock that exist for AI, there has to be a complete perspective on what are the risks, limitations, and safety considerations that could exist for AMI. Winkler et. al. (2025) looks at these issues such as training data bias, innapropriate prompts during testing and operation, black-box nature and flawed reward-system training (“Key risks and limitations of AI systems” section). There is a discussion on how “distributional shift” in an AI can cause hallucinations as biased training data can shift the AI to a innapropriate or incorrect response, similarly innapropate prompts that are outside of the scope of the AI can cause these hallucinations (Winkler et. al., 2025, “Distributional Shift” section). This is why extensive testing of the datasets and the AI have to be done to ensure the training data is not biased and that innapropriate prompts are caught and informed to the user. The way that the AI produces responses is considered as black-box because for a non-developer of the product, it is difficult to understand all the decisions that the AI makes to create the response (Winkler et. al., 2025, “Black box decision-making” section). This is why AMI intends to be one of the stepping stones to fix this problem by providing the sources/references used to generate the response. One of the ways that AIs or ML (Machine Learning) algorithms are taught is reinforcement learning that uses positive and negative feedback, the issue is that the AI can try to avoid negative feedback by trying to generalize a prompt to avoid mistakes (Winkler et. Al., 2025, “Reinforcement learning” section). By doing this, the AI may misinterpret a prompt or create false predictions, which come from training data that is “insufficient, unaligned, or poor-quality” (Winkler et. Al., 2025, “Reinforcement learning” section). All of these concerns need to be addressed when designing, implementing, and testing AMI.

Analysis of open-source AI virtual assistants in the medical field

MedGemma is from Google, a transformer-based LLM that provides medical Q&A, healthcare reasoning, and medical language understanding. In Google DeepMind, the current demos available are: EHR Navigator Agent, Appoint-Ready, Radiology Explainer, Radiology Learning Companion (<https://deepmind.google/models/gemma/medgemma>).

BioMistral-7B is from Mistral AI that inherits much of the base Mistral-7B model architecture for biomedical text understanding, scientific Q&A, and clinical reasoning. A comprehensive review of BioMistral-7B can be found in GitHub (<https://github.com/Incredible88/BioMistral-Clinical-7B>).

LLaVA-Med is from Microsoft that is based on the LLaVA architecture, with it being a LMM (Large Multimodal Model) that focuses on medical image generation (as an input and output) that supports conversational text prompts. A comprehensive review of LLaVA-Med can be found in GitHub (<https://github.com/microsoft/LLaVA-Med>).

Conclusion

Analysis of these open-source AI models showed promise that AMIs goals can be designed and developed. As AI continues to be integrated in the medical field, it becomes increasingly apparent that transparency to all users whether it be doctors, medical professionals or patients, which part of that solution can be source transparency.

Bibliography

Bajwa, J., Munir, U., Nori, A., & Williams, B. (2021). Artificial Intelligence in healthcare:

Transforming the practice of medicine. *Future Healthcare Journal*, 8(2), e188-e194. doi:

<https://doi.org/10.7861/fhj.2021-0095>

(Found in the FSU library database) A comprehensive review of how AI was being used and developed during pandemic, shows a predictive roadmap on the expectation on how AI was going to be implemented into healthcare. This will provide background information on how AI initially started transforming the medical space.

Magnini, M., Aguzzi, G., & Montagna, S. (2025). Open-source small language models for

personal medical assistant chatbots. *Intelligence-Based Medicine*, 11, Article 100197.

doi: <https://doi.org/10.1016/j.ibmed.2024.100197>

(Found in the FSU library database) Reviews the existence of privacy concerns in AI LMs and the performance of locally run open-source LMs compared to Gemini Pro 1.5 which is made as a benchmark. It provides the necessary perspective on how AMI is supposed to address not just the hallucination concerns but also to ensure that privacy and safety through guardrails is just as important.

Tanasă, A., Opera, S., & Bâra, A. (2026). Designing an architecture of a multi-agentic AI-

powered virtual assistant using LLMs and RAG for a medical clinic. *Electronics*, 15(2),

Article 334. doi: <https://doi.org/10.3390/electronics15020334>

(Found in the FSU library database) This source is currently unused due to time constraints.

Winkler, P. W., Zsidai, B., Senorski, E. H., Pruneski, J. A., Hirschmann, M. T., Ley, C., Tischler, T., Herbst, E., Pareek, A., Musahl, V., Oeding, J. F., Oetl, F. C., Longo, U. G., Samuelsson, K., Feldt, R., & ESSKA Artificial Intelligence Working Group (2025). A practical guide to the implementation of AI in orthopaedic research-Part 7: Risks, limitations, safety and verification of medical AI systems. *Journal of Experimental Orthopaedics*, 12(2), e70247. doi: <https://doi.org/10.1002/jeo2.70247>

(Found in the FSU library database) Looks at the challenges (risks and limitations) of AI in healthcare and how they are being addressed through considerations, regulations and frameworks. This provides the context needed to understand what potential challenges exist when designing and developing AMI and how to combat them.